# TechXpress

E-commerce Platform with ASP.NET Core
Requirements Gathering

Full Stack .NET Project 1 | 2 Oct - 30 Nov 2026 | Team: 5 members

| | | | |
|---|---|---|---|
| Start date | Friday, 2 October 2026 | Delivery date | Monday, 30 November 2026 |
| Requirements baseline | Thursday, 15 October 2026 (M2) | Document owner | Aya Kamel (assembly); all members contribute |
| Project type | E-commerce web app | Architecture | N-layer + Repository + Unit of Work |

## 1. Introduction

**Purpose.** This document records what TechXpress must do and how well it must do it, before and while the team builds it. It is the single reference for scope, priorities and acceptance, and it feeds the test plan and the project plan.

**Sources.** Requirements marked **Brief** come directly from the course file Full Stack .NET Web Developer (Project 1). Requirements marked **Proposed** were added by the team to make the system complete and safe to use; they must be confirmed with the course instructor before the baseline on 15 October.

**Priority scale (MoSCoW). Must**: the project fails without it. **Should**: important, cut only under pressure. **Could**: nice to have, first to be dropped (see Plan B in the project plan).

### 1.1 Scope summary

| In scope | Out of scope (unless the team agrees) |
|---|---|
| • Product catalog with search, filters, pagination<br>• Accounts, login and roles (Customer / Admin)<br>• Session-based cart with discounts<br>• Checkout and Stripe payment (test mode)<br>• Customer profile and order history<br>• Admin dashboard (DataTables)<br>• Deployment and documentation | • Mobile app<br>• Product ratings and reviews<br>• Multi-currency / multi-language<br>• Courier shipping integration |

### 1.2 Glossary

| Term | Meaning |
|---|---|
| Customer | A signed-in shopper who can buy products and see their own orders |
| Admin | A store employee who manages products, categories and orders |
| Guest | A visitor who is not signed in; can browse and use a cart |
| Webhook | A message Stripe sends to the site when a payment event happens |
| Unit of Work | A class that groups repository work so it is saved or rolled back together |
| RBAC | Role-based access control: what a user may open depends on the role |

# 2. Stakeholders and Actors

| Stakeholder | Role in the project | Main interest | How we involve them |
|---|---|---|---|
| Course instructor / evaluator | Primary stakeholder; approves scope and grades the result | Brief followed, architecture patterns used, working deployed site | Confirm Proposed requirements; baseline review on 15 Oct |
| Customers (shoppers) | End users of the store | Easy search, safe payment, clear order status | Personas and user stories; usability checks in joint testing |
| Store admin | Runs the catalog and orders | Fast product and order management | Admin user stories; walkthrough of the dashboard |
| Project team (5 members) | Build and test the system | Clear scope, fair split of backend work | Workshops, reviews, change control |
| Stripe (external system) | Payment provider | Correct use of Checkout and webhooks | Follow Stripe documentation; test mode only |
| Hosting provider (Azure / Hostinger) | Runs the deployed site | Valid configuration and settings | Deployment checklist in phase 4 |

## 2.1 User roles

| Role | Can do | Cannot do |
|---|---|---|
| Guest | Browse, search, filter, view details, use the cart, register, log in | Check out, see orders, open admin pages |
| Customer | Everything a guest can, plus checkout, pay, edit profile, see own order history | See other customers' orders, open admin pages |
| Admin | Manage products, categories and orders; view the dashboard | Pay on behalf of customers or edit payment records |

# 3. Requirements Gathering Approach

Requirements are gathered by area. The person who owns an area leads the work for it, so every member contributes, and the backend areas are shared across the team.

## 3.1 Techniques

| Technique | Used for | Result |
|---|---|---|
| Document analysis | Reading the course brief line by line | List of features, technologies and deliverables |
| Team workshops | Scope, actors, priorities (MoSCoW) | Scope statement and prioritized list |
| Competitor review | Comparing three electronics stores' catalog, cart and checkout | Feature comparison notes, ideas for Proposed items |
| Instructor questions | Confirming the open questions in section 9 | Confirmed or rejected Proposed items |
| Wireframes | Catalog, cart, checkout, admin screens | Screen sketches used as acceptance reference |
| User stories | Writing needs from the user's point of view | Stories with acceptance criteria (section 5) |
| Review and sign-off | Checking the whole list for gaps and conflicts | Baseline version 1.0 |

## 3.2 Activity plan

| Activity | Technique | Owner | Dates | Output |
|---|---|---|---|---|
| Analyze the brief; extract features and constraints | Document analysis | All | 2 - 3 Oct | Feature list |
| Identify stakeholders and actors | Workshop | Mostafa, Aya Kamel | 2 - 3 Oct | Section 2 |
| Scope workshop: in scope and out of scope | Team workshop | All | 5 Oct (M1) | Scope statement, draft v0.1 |
| Payment and checkout requirements | Stripe docs study | Aya Tarek | 2 - 8 Oct | PAY and ORD requirements |
| Competitor review of three stores | Competitive analysis | Doaa, Aya Tarek | 6 - 8 Oct | Comparison notes |
| Account, role and security requirements | Checklist, instructor questions | Ahmed | 6 - 9 Oct | AUTH, PRF, NFR-01 |
| Data, architecture and transaction constraints | Technical review | Mostafa | 6 - 9 Oct | TEC, NFR-02 |
| Admin needs and acceptance criteria | Wireframes, user stories | Aya Kamel | 6 - 12 Oct | ADM, acceptance criteria |
| Catalog and cart wireframes | Prototyping | Doaa, Aya Tarek | 9 - 12 Oct | Wireframes |
| Review and MoSCoW prioritization | Review meeting | All | 12 - 14 Oct | Prioritized list |
| Baseline sign-off (M2) | Approval | All + instructor | 15 Oct | Baseline v1.0 |
| Change control | Change requests | All | 16 Oct - 26 Nov | Change log until code freeze |

# 4. Functional Requirements

Source: **Brief** = stated in the course file; **Proposed** = added by the team, to confirm with the instructor.

## 3.1 Authentication and accounts (AUTH) - owner: Ahmed

| ID | Requirement | Priority | Source | Owner |
|---|---|---|---|---|
| FR-AUTH-01 | A visitor can register with name, email and password. | Must | Brief | Ahmed |
| FR-AUTH-02 | A registered user can log in and log out. | Must | Brief | Ahmed |
| FR-AUTH-03 | Two roles exist, Customer and Admin; every new account becomes a Customer. | Must | Brief | Ahmed |
| FR-AUTH-04 | Admin-only pages reject other users and show an AccessDenied page. | Must | Brief | Ahmed |
| FR-AUTH-05 | A default Admin account is seeded so the dashboard can be reached after deployment. | Must | Proposed | Ahmed |
| FR-AUTH-06 | Password rules and temporary lockout after repeated failed logins are enforced. | Should | Proposed | Ahmed |
| FR-AUTH-07 | The navigation bar shows items according to sign-in state and role. | Should | Proposed | Ahmed |
| FR-AUTH-08 | After login the user returns to the page they originally requested. | Could | Proposed | Ahmed |

## 3.2 Product catalog (CAT) - owner: Doaa

| ID | Requirement | Priority | Source | Owner |
|---|---|---|---|---|
| FR-CAT-01 | The product list shows image, name, price and category. | Must | Brief | Doaa |
| FR-CAT-02 | Customers can search products by name. | Must | Brief | Doaa |
| FR-CAT-03 | Customers can filter by category and price range. | Must | Brief | Doaa |
| FR-CAT-04 | Product lists are paginated. | Must | Brief | Doaa |
| FR-CAT-05 | A product details page shows images, description, specifications and stock status. | Must | Brief | Doaa |
| FR-CAT-06 | Live search suggestions appear while typing (jQuery / AJAX). | Should | Brief | Doaa |
| FR-CAT-07 | Discounted price and stock status are shown on product cards. | Should | Brief | Doaa |
| FR-CAT-08 | Customers can sort by price and by newest. | Should | Proposed | Doaa |
| FR-CAT-09 | A home page shows latest products and categories. | Should | Proposed | Doaa |
| FR-CAT-10 | The details page shows related products from the same category. | Could | Proposed | Doaa |

## 3.3 Shopping cart (CRT) - owner: Aya Tarek

| ID | Requirement | Priority | Source | Owner |
|---|---|---|---|---|
| FR-CRT-01 | A customer can add a product to the cart. | Must | Brief | Aya Tarek |
| FR-CRT-02 | A customer can change quantity, remove an item and clear the cart. | Must | Brief | Aya Tarek |
| FR-CRT-03 | The cart keeps its contents during the browsing session (Session State). | Must | Brief | Aya Tarek |
| FR-CRT-04 | Totals and discounts are calculated in the business (Services) layer. | Must | Brief | Aya Tarek |
| FR-CRT-05 | A quantity above available stock is rejected with a clear message. | Must | Proposed | Aya Tarek |
| FR-CRT-06 | The Navbar shows the number of items in the cart. | Should | Proposed | Aya Tarek |
| FR-CRT-07 | A guest can fill a cart; login is required at checkout. | Should | Proposed | Aya Tarek |

### 3.4 Orders and checkout (ORD) - owner: Aya Tarek, with Mostafa

| ID | Requirement | Priority | Source | Owner |
|---|---|---|---|---|
| FR-ORD-01 | Checkout collects shipping details and shows an order summary. | Must | Brief | Aya Tarek |
| FR-ORD-02 | The order and its lines are saved in one transaction through the Unit of Work. | Must | Brief | Aya Tarek |
| FR-ORD-03 | An order moves through the statuses Pending, Paid, Shipped, Delivered or Cancelled. | Must | Proposed | Aya Tarek |
| FR-ORD-04 | Order lines store the unit price at purchase time. | Should | Proposed | Aya Tarek |
| FR-ORD-05 | Each order has a unique, human-readable order number. | Should | Proposed | Aya Tarek |

### 3.5 Online payment (PAY) - owner: Aya Tarek

| ID | Requirement | Priority | Source | Owner |
|---|---|---|---|---|
| FR-PAY-01 | The customer pays through Stripe Checkout (test mode). | Must | Brief | Aya Tarek |
| FR-PAY-02 | A Stripe webhook confirms payment and moves the order from Pending to Paid. | Must | Brief | Aya Tarek |
| FR-PAY-03 | Transaction data (payment reference, amount, status) is stored with the order. | Must | Brief | Aya Tarek |
| FR-PAY-04 | The webhook verifies the Stripe signature and is safe to receive twice. | Must | Proposed | Aya Tarek |
| FR-PAY-05 | Cancelled or declined payment leaves the order unpaid and the stock unchanged. | Must | Proposed | Aya Tarek |
| FR-PAY-06 | Stripe keys are kept in User Secrets and Azure App Settings, never in source control. | Must | Proposed | Aya Tarek |

### 3.6 Profile and order history (PRF) - owner: Ahmed

| ID | Requirement | Priority | Source | Owner |
|---|---|---|---|---|
| FR-PRF-01 | A customer can view and edit profile data (name, phone, address). | Must | Brief | Ahmed |
| FR-PRF-02 | A customer can see a list of their past orders and open an order's details. | Must | Brief | Ahmed |
| FR-PRF-03 | A customer can only see their own orders. | Must | Proposed | Ahmed |
| FR-PRF-04 | A customer can change their password. | Should | Brief | Ahmed |

### 3.7 Admin dashboard (ADM) - owner: Aya Kamel

| ID | Requirement | Priority | Source | Owner |
|---|---|---|---|---|
| FR-ADM-01 | Admins manage products in a DataTables grid with sorting, filtering and paging. | Must | Brief | Aya Kamel |
| FR-ADM-02 | Admins can add, edit and delete products, including image upload. | Must | Brief | Aya Kamel |
| FR-ADM-03 | Admins can create, edit and delete categories. | Must | Brief | Aya Kamel |
| FR-ADM-04 | Admins can list orders, filter by status and date, and open order details. | Must | Brief | Aya Kamel |
| FR-ADM-05 | Admins can change an order's status. | Must | Proposed | Aya Kamel |
| FR-ADM-06 | Success and error messages use Toastr; deletes ask for confirmation. | Should | Brief | Aya Kamel |
| FR-ADM-07 | A dashboard shows counts of products and orders, sales total and latest orders. | Should | Proposed | Aya Kamel |
| FR-ADM-08 | Admins can view users and assign roles. | Could | Proposed | Aya Kamel |

### 3.8 Data and technical requirements (TEC) - owner: Mostafa

| ID | Requirement | Priority | Source | Owner |
|---|---|---|---|---|
| FR-TEC-01 | The solution has Web, Services and Data projects with one-way references. | Must | Brief | Mostafa |
| FR-TEC-02 | A generic repository plus specific repositories hide EF Core from the Services layer. | Must | Brief | Mostafa |
| FR-TEC-03 | A Unit of Work coordinates repositories and supports transactions. | Must | Brief | Mostafa |
| FR-TEC-04 | All services and repositories are registered through Dependency Injection. | Must | Brief | Mostafa |
| FR-TEC-05 | Database changes are applied through EF Core Migrations created only by Mostafa. | Must | Proposed | Mostafa |
| FR-TEC-06 | Seed data provides categories and 15 - 20 sample products for demos and tests. | Must | Proposed | Mostafa |

### 4.1 Summary

| Total | Must | Should | Could | From the brief | Proposed |
|---|---|---|---|---|---|
| 54 | 38 | 13 | 3 | 32 | 22 |

## 5. User Stories and Acceptance Criteria

| ID | Story | Acceptance criteria | Covers |
|---|---|---|---|
| US-01 | As a visitor, I want to register and log in, so that I can buy and see my orders. | Given I enter a new email and a valid password, when I submit, then my account is created as Customer. Given an email that exists, then I see a clear error. | FR-AUTH-01, 02, 03 |
| US-02 | As a shopper, I want to search and filter products, so that I find what I need quickly. | Given a search term, a category and a price range, when I apply them together, then only matching products appear with pagination. | FR-CAT-02, 03, 04 |
| US-03 | As a shopper, I want live suggestions while I type, so that I reach a product faster. | Given I type at least two characters, when I pause, then matching product names appear without reloading the page. | FR-CAT-06 |
| US-04 | As a shopper, I want a cart that updates my total, so that I know what I will pay. | Given items in the cart, when I change a quantity or remove an item, then the total, including discounts, updates at once. A quantity above stock is refused. | FR-CRT-01 to 05 |
| US-05 | As a customer, I want to pay securely online, so that my order is confirmed. | Given a filled cart, when I pay with a Stripe test card, then I return to a success page and the order shows Paid after the webhook arrives. | FR-ORD-01, 02, FR-PAY-01 to 04 |
| US-06 | As a customer, I want a declined payment to leave no side effects, so that I am not charged for goods I did not get. | Given a declined card or a cancelled payment, then the order stays unpaid and the product stock is unchanged. | FR-PAY-05 |
| US-07 | As a customer, I want to see my past orders, so that I can track what I bought. | Given I am signed in, when I open order history, then I see only my orders; opening another customer's order is blocked. | FR-PRF-02, 03 |
| US-08 | As an admin, I want to manage products in a table, so that the catalog stays current. | Given the admin grid, when I add, edit or delete a product, then the table updates at once and a Toastr message confirms it. Deletes ask for confirmation. | FR-ADM-01, 02, 06 |
| US-09 | As an admin, I want to update order statuses, so that customers see progress. | Given an order, when I change its status, then the new status is saved and shown in the customer's order history. | FR-ADM-04, 05 |
| US-10 | As the owner, I want customers kept out of admin pages, so that store data is protected. | Given a signed-in Customer, when they open an admin URL, then they are sent to AccessDenied. | FR-AUTH-04 |

## 6. Non-Functional Requirements

| ID | Area | Requirement | Owner |
|---|---|---|---|
| NFR-01 | Security | Production runs on HTTPS; passwords are hashed by Identity; every admin endpoint is authorized; all POST actions use anti-forgery tokens; secrets stay out of the repository. | Ahmed |
| NFR-02 | Data integrity | No partial orders: order and lines are saved in one transaction with rollback on failure; the webhook can be received twice without double effect. | Mostafa, Aya Tarek |
| NFR-03 | Performance | Proposed target: the product list answers in about 2 seconds on the deployed site with seed data, using paging and read-only queries. | Mostafa, Doaa |
| NFR-04 | Usability | The site works on phone, tablet and desktop using Bootstrap; layout, colors and messages are consistent across pages. | Doaa |
| NFR-05 | Maintainability | Three layers, Dependency Injection, clear naming, and Pull Request review before every merge to develop. | Mostafa |
| NFR-06 | Testability | Each member writes xUnit tests for the main paths of their own services; the test plan is executed before the 26 Nov code freeze. | Aya Kamel |
| NFR-07 | Compatibility | Proposed: works in current versions of Chrome, Edge and Firefox. | Doaa |
| NFR-08 | Deployability | The site is live on Azure or Hostinger by 30 Nov 2026; migrations and settings are documented so the deployment can be repeated. | Aya Kamel, Mostafa |
| NFR-09 | Privacy | The session cookie that holds the cart is configured deliberately (marked essential) and cookie use is described in the documentation. | Aya Tarek |
| NFR-10 | Documentation | Architecture explanation, setup guide and user manual are delivered with the project; each member documents their own slice. | Aya Kamel, all |

## 7. Business Rules

| ID | Rule | Source |
|---|---|---|
| BR-01 | A product can be bought only while its stock is at least the requested quantity. | Proposed |
| BR-02 | Cart total = sum of (discounted unit price x quantity); discounts are calculated in the Services layer, not in views. | Brief |
| BR-03 | An order and all its lines are saved together or not at all. | Brief |
| BR-04 | Stock is reduced once, when the Stripe webhook confirms payment; a repeated webhook has no further effect. | Proposed |
| BR-05 | Order statuses: Pending (created), Paid (webhook), Shipped, Delivered, Cancelled. Only an Admin can set Shipped, Delivered or Cancelled. | Proposed |
| BR-06 | Order lines keep the unit price paid, so later price changes do not alter past orders. | Proposed |
| BR-07 | A customer can read only their own orders; an Admin can read all orders. | Proposed |
| BR-08 | Only Admins can create, edit or delete products and categories. | Brief |
| BR-09 | A category that still has products cannot be deleted. | Proposed |
| BR-10 | Checkout requires a signed-in Customer; a guest cart is kept and used after login. | Proposed |

## 8. Data Requirements

Main entities that the Data layer must support. Mostafa creates and migrates them; field details are finalized when the DbContext is built (M1, 5 Oct).

| Entity | Key attributes | Relationships |
|---|---|---|
| Category | Id, Name, Description | One category has many products |
| Product | Id, Name, Description, Price, DiscountPercent, Stock, ImageUrl, CategoryId, CreatedAt | Belongs to one category; appears in many order lines |
| ApplicationUser | Id, Email, FullName, Phone, Address (extends Identity user) | Has many orders; has one or more roles |

| Entity | Key attributes | Relationships |
|---|---|---|
| Order | Id, OrderNumber, UserId, Status, Total, ShippingDetails, CreatedAt | Belongs to one user; has many order lines; has payment data |
| OrderDetail | Id, OrderId, ProductId, Quantity, UnitPrice | Belongs to one order and one product |
| Payment data | Stripe session / payment reference, Amount, Status, PaidAt | Linked to one order |

# 9. Constraints, Assumptions and Open Questions

## 9.1 Constraints

- The stack is fixed by the brief: ASP.NET Core MVC, EF Core, Identity, Stripe, jQuery, DataTables, Toastr, Bootstrap.
- The architecture must use layers, the Repository pattern, Unit of Work and Dependency Injection.
- Fixed dates: start 2 Oct 2026, baseline 15 Oct, code freeze 26 Nov, delivery 30 Nov 2026.
- Team of five; only Mostafa creates Migrations.
- Stripe runs in test mode; no real payments.

## 9.2 Assumptions

- One currency and one language (English) for the site.
- Product images are stored on the server and referenced by path.
- A single hosted instance is enough, so the Session cart is acceptable.
- Azure or Hostinger access is available to the team by 12 Nov.

## 9.3 Open questions for the instructor

| # | Question | Affects | Needed by |
|---|---|---|---|
| 1 | Which currency, tax and shipping-fee rules should the checkout use? | ORD, PAY | 15 Oct |
| 2 | Is guest checkout allowed, or must customers register first? | CRT, ORD | 15 Oct |
| 3 | Are discounts set per product, or are coupon codes expected? | CRT, ADM | 15 Oct |
| 4 | Azure or Hostinger for the final deployment? | NFR-08 | 12 Nov |
| 5 | Is an Arabic / right-to-left interface required? | CAT, NFR-04 | 15 Oct |
| 6 | Are email notifications (order confirmation) expected? | ORD | 15 Oct |
| 7 | Which browsers must be supported in the demonstration? | NFR-07 | 15 Oct |

# 10. Traceability

Each requirement group is traced to the layers that implement it, the project phase and owner, and the test area that checks it.

| Group | Layers | Phase and dates | Owner | Test area |
|---|---|---|---|---|
| AUTH, PRF | Data, Services, Web | Phase 1 - 3 (2 Oct - 12 Nov) | Ahmed | Registration, Login, Permissions, Profile |
| CAT | Data, Services, Web | Phase 1 - 3 (2 Oct - 12 Nov) | Doaa | Products |
| CRT | Services, Web | Phase 2 (16 - 29 Oct) | Aya Tarek | Cart |
| ORD, PAY | Data, Services, Web | Phase 3 (30 Oct - 12 Nov) | Aya Tarek | Order, Stripe |
| ADM | Data, Services, Web | Phase 2 - 3 (16 Oct - 12 Nov) | Aya Kamel | Admin |
| TEC | Data, composition root | Phase 1 - 3 (2 Oct - 12 Nov) | Mostafa | Unit tests, Deployed site |
| NFR-01 to 10 | All layers | Phase 1 - 4 (2 Oct - 30 Nov) | As listed in section 6 | Security review, Full testing |

# 11. Requirements Management

## 11.1 Change control

- After the baseline on 15 October, any change is written as a change request: what, why, which requirements and tasks it touches, and its effect on the dates.
- The weekly review decides each request: accept, defer to Could, or reject. Accepted changes update this document and the project plan together.
- Between 26 November (code freeze) and 30 November only bug fixes are allowed.
- Every change is recorded in a change log with its date, requester and decision.

## 11.2 Definition of a complete requirement

- It has an ID, a priority, a source and one owner.
- It can be tested: someone can say pass or fail.
- It does not conflict with another requirement or with a business rule.
- It appears in the traceability table and in the test plan.

## 11.3 Review and sign-off

| Name | Role | Responsible for | Signature | Date |
|---|---|---|---|---|
| Mostafa Soliman Mohamed Salem | Tech Lead / Data Layer | TEC, data requirements | | |
| Ahmed Hassan Ali Abotayesh | Auth, Profile & Security | AUTH, PRF, NFR-01 | | |
| Doaa Ali Sayed Omar | Catalog & UI | CAT, NFR-04 | | |
| Aya Tarek Maghawry Elhwary | Cart, Orders & Payment | CRT, ORD, PAY | | |
| Aya Kamel Soliman | Admin, QA & DevOps | ADM, document assembly, NFR-06 | | |
| Course instructor | Approver | Baseline approval | | |